The Incremental Value of Player Information in Football Match Prediction

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Abstract

Closing bookmaker odds provide a benchmark for football forecasting because they summarize information available shortly before kick-off. Yet changing line-ups and fluctuations in individual form could contain information that market probabilities do not fully reflect. This study therefore examined whether the recent recorded performances of identified starters improved three-way outcome predictions beyond a recalibrated closing market.

The analysis used 7,801 eligible top-division matches from Belgium, the Netherlands, Portugal, Scotland, and Turkey. A chronological development sample contained 6,638 matches from 2020/21–2024/25, while 1,163 matches from 2025/26 formed a final held-out test. Player and team histories were constructed from earlier matches. All feature choices, model settings, comparisons, and the decision not to apply post-hoc calibration were fixed before the final season was examined. The primary comparison placed a pooled market-only multinomial logistic model against an otherwise identical model augmented by four recent player-performance features. Each league received equal importance, and performance was assessed primarily through log loss, supported by Brier score, ranked probability score, country-level estimates, and paired match-week bootstrap intervals.

On the held-out season, the player-enhanced model recorded an equal-league log loss of 0.9580, compared with 0.9567 for the recalibrated market: a 0.143% deterioration. The paired 95% interval ranged from a 0.511% deterioration to a 0.234% improvement. Brier score and ranked probability score followed the same pattern, while country-level effects were small, mixed, and uncertain. These findings show no reliable incremental value from the tested player features and emphasize that player-enhanced forecasts require evaluation against strong market-only baselines on unseen data.

1 Introduction

Probabilistic football forecasting is used to compare teams, support sporting decisions, and estimate home-win, draw, and away-win probabilities. Match outcomes depend on team quality, tactics, venue, scheduling, and the players available, so useful forecasts must assign credible probabilities to all three outcomes rather than only identify the most likely result.

Bookmaker closing odds provide a particularly strong benchmark for this task. After removing the bookmaker margin, they summarize information available to market participants shortly before kick-off, including team strength, injuries, line-ups, and recent results. Outperforming a historical or team-strength baseline therefore does not establish that a new input adds information beyond the market.

Player-level information is a plausible candidate for such an improvement. Changing line-ups may make recent starter performance more responsive than team histories to transfers, injuries, and rotation. However, a fitted model can also appear to improve raw odds merely by correcting systematic market bias. The relevant test is therefore an otherwise identical comparison between a recalibrated market-only model and one augmented with player information.

The study asks:

> *Does the recent recorded performance of identified starting players improve three-way football match predictions beyond recalibrated closing bookmaker probabilities?*

The empirical analysis covers the top divisions of Belgium, the Netherlands, Portugal, Scotland, and Turkey. Player and team histories were updated chronologically within each league. Feature selection and model development used 2020/21–2024/25; all choices were then fixed before a single evaluation on 2025/26. The primary pooled comparison gave each league equal weight and was assessed by log loss, supported by Brier score, ranked probability score, country-level estimates, and paired match-week uncertainty intervals.

The contribution is both substantive and methodological. The study isolates the incremental value of recent starter information under identical market recalibration capacity, and it does so with an auditable separation between development and final testing. The held-out results provide no reliable evidence that the tested player summaries improved the recalibrated closing market.

2 Related Literature

2.1 Football Match Prediction

Statistical football forecasting has traditionally represented a team through its previous results and scoring record. One foundational approach models the goals scored by each side as Poisson variables whose rates depend on the teams' attacking and defensive strengths (Maher, 1982). This framework produces a full score distribution and, by aggregation, probabilities for a home win, draw, and away win. Dixon and Coles (1997) extended it by correcting the probabilities of low-scoring results and weighting recent observations more heavily. The resulting Dixon–Coles model remains an influential, interpretable benchmark that adapts as results are observed.

Later work developed richer representations of changing team strength. Rue and Salvesen (2000) treated team abilities as dynamic quantities in a Bayesian model, while Koopman and Lit (2015) allowed attacking and defensive strengths to evolve within a dynamic bivariate Poisson system. Other studies modelled the three-way outcome more directly. Goddard (2005), for example, compared regression models for goals and match results, and Hvattum and Arntzen (2010) used Elo ratings as inputs to match-result forecasts. Despite different assumptions, each summarizes history at team level.

Machine-learning research has widened the set of relationships that can be estimated from such data. Using English Premier League matches, Baboota and Kaur (2019) combined engineered performance features with several classification methods. Such models can capture nonlinearities and interactions, but their flexibility increases the choices made during development. More fundamentally, success against an outcome-frequency, Elo, or Poisson benchmark does not show that a model contains information absent from bookmaker prices. For the present research question, team-based models are therefore useful comparators, but the closing market is the more demanding reference point.

2.2 Bookmaker Markets as a Forecasting Benchmark

Fixed odds can be interpreted as probabilistic forecasts after accounting for the bookmaker's margin. Their forecasting strength has been demonstrated in direct comparisons with statistical models. In almost 10,000 English matches, Forrest et al. (2005) found that a model containing a broad set of observable variables did not outperform the forecasts embedded in bookmaker odds. Štrumbelj and Robnik-Šikonja (2010) reached a similarly strong assessment across six European leagues, while also showing that forecast quality differed between bookmakers and competitions and improved over time. These findings support the use of market probabilities as a benchmark without requiring the stronger claim that every quoted price is perfectly efficient.

Indeed, the literature contains evidence of both information efficiency and small, context-dependent departures from it. Dixon and Pope (2004) examined whether statistical forecasts retained value in the UK football betting market, whereas Angelini and De Angelis (2019) found differing degrees of efficiency across European online football markets. Market structure also matters: Franck et al. (2010) reported that betting-exchange prices were more accurate than the corresponding forecasts from bookmakers in their sample. Collectively, these studies suggest that odds are difficult to improve upon, but that their quality can depend on the provider, competition, and point at which they are observed. This motivates the use of the latest consistently available closing odds in the present study.

Raw inverse odds are not probabilities because their sum normally exceeds one. Removing this overround is therefore part of the forecasting problem rather than a purely cosmetic transformation. Štrumbelj (2014) showed that alternative methods for recovering probabilities from odds can produce different calibration and forecasting properties. A second distinction is equally important here. A trained model supplied with bookmaker probabilities may improve their log loss simply by learning systematic recalibration, even if every additional football feature is uninformative. Consequently, a market-plus-player model must be compared with a market-only model that receives the same opportunity to recalibrate the odds. Comparing only with raw probabilities would not identify whether an apparent gain came from player information or from recalibration.

2.3 Player-Level Information

Team-level histories treat the club as the main forecasting unit, even though its strength depends on the players selected. Player research has approached this problem from two broad directions. Bottom-up systems summarize recorded actions, while top-down systems infer a player's underlying contribution from the results produced while that player is on the pitch. An early example of the first direction is the event-based player performance rating developed by McHale et al. (2012). In the second, plus–minus models distribute changes in team performance across the participating players while adjusting for teammates and opponents. Kharrat et al. (2020) developed goal-, expected-goal-, and expected-points-based versions of such ratings for football.

Starting line-ups provide a natural way to translate player estimates into team forecasts. Rather than estimating a separate fixed strength for every team, Maystre et al. (2016) related matches through the players appearing in their line-ups, allowing information to be shared across club and international competitions. Arntzen and Hvattum (2021) compared team Elo ratings with aggregated starting-player plus–minus ratings. The two sources performed similarly when used separately, but forecasts combining them were significantly better than forecasts based on either source alone. This result demonstrates that player and team ratings can contain complementary information, although it does not by itself establish improvement beyond the betting market.

The closest precedent for that stronger comparison is Holmes and McHale (2024), who modelled changing player ability, aggregated the selected players into team strength, and evaluated match-result forecasts against bookmaker predictions. Their approach treats ability as a latent quantity that evolves through time. The present study asks a narrower but distinct question: whether recent directly recorded performance from the starting players adds value. That distinction matters because observed actions may be more transparent than a latent rating but also substantially noisier. Consistent with this concern, Gelade and Hvattum (2020) found that event-level statistics explained part of the variation in plus–minus ratings but produced only a marginal improvement in their match-prediction power.

Recent form also depends on how performance history is summarized. He et al. (2024) compared short rolling histories when forecasting individual match actions and showed that predictability varied with the action, playing position, and amount of recent information used. This supports treating window length as a modelling choice rather than assuming that more history is always better. It also highlights a limitation of rolling observed performance: such features mix persistent player quality with temporary form. The features tested in this paper deliberately represent recent recorded output; they are not presented as estimates of a player's underlying ability.

2.4 Research Gap

Previous work establishes that team-strength models are useful, bookmaker odds are formidable forecasts, and player ratings can improve models built only from team information. It does not fully resolve whether short-term, recorded player performance adds information beyond a recalibrated closing market. The closest studies generally estimate latent player ability, focus on major competitions, or answer whether player ratings complement team ratings rather than whether recent player output complements closing odds.

Isolating that contribution requires chronological features, identical matches, equal recalibration opportunities, and a test period not used for feature or model selection. In a multi-league study, pooled evidence should also avoid being dominated by the largest competition and should report country-level uncertainty. The design below incorporates these requirements in a common five-league sample.

3 Data

3.1 Countries and Seasons

The study covered the national top divisions of Belgium, the Netherlands, Portugal, Scotland, and Turkey over six seasons from 2020/21 to 2025/26. These competitions were processed under one data specification. The first five seasons served as the development and training period. The 2025/26 season was stored separately until the full modelling specification had been fixed.

Greece was collected and audited but excluded from the modelling sample before the comparative experiments because its first two seasons lacked enough complete starting line-ups for the common chronological folds. The exclusion followed the declared coverage rule rather than model performance. Regular-season, Scottish post-split, and applicable top-division play-off fixtures were retained when eligible.

3.2 Match and Odds Data

Match schedules, dates, teams, completion status, scores, line-ups, and player-match records were obtained from TheStatsAPI (TheStatsAPI, nd). Historical three-way odds were obtained from Football-Data.co.uk (Football-Data.co.uk, nd). Records were joined within league and season using match date and normalized team identities. A match entered the sample only when the final home and away scores agreed across both sources.

Only rows explicitly identified in the odds data as closing prices were used. All three decimal odds had to be present and greater than one. The bookmaker margin was removed by taking the inverse of each price and dividing it by the sum of the three inverse prices. This produced positive home, draw, and away probabilities that summed to one. They were treated solely as forecasts; the study did not simulate betting returns.

3.3 Player and Starting-Line-Up Data

The player records contained a stable player identifier, team assignment, position, starting status, minutes played, and event-level match statistics. These included goals, assists, shots, key passes, tackles, interceptions, saves, and provider ratings. Expected-goal fields were audited separately, but non-penalty expected goals were not used in the final active feature set. The final prediction sample required exactly eleven starters for the home team and eleven for the away team. Every starter required a populated identifier and recorded minutes, and every player's team identifier had to correspond to one of the two clubs in the fixture.

Player histories were formed independently inside each league. Under the primary rule, a player entering a new country contributed only appearances recorded in the current league; earlier performances from a foreign league were not carried across competitions. This avoided imposing an unestimated conversion between different league standards. Matches that lacked odds or otherwise failed as prediction targets could still contribute an earlier player appearance when a valid player record existed. Thus, excluding a match from forecast evaluation did not erase information available before a later match. Historical-variable construction is described in Section 4.

3.4 Inclusion Criteria and Construction

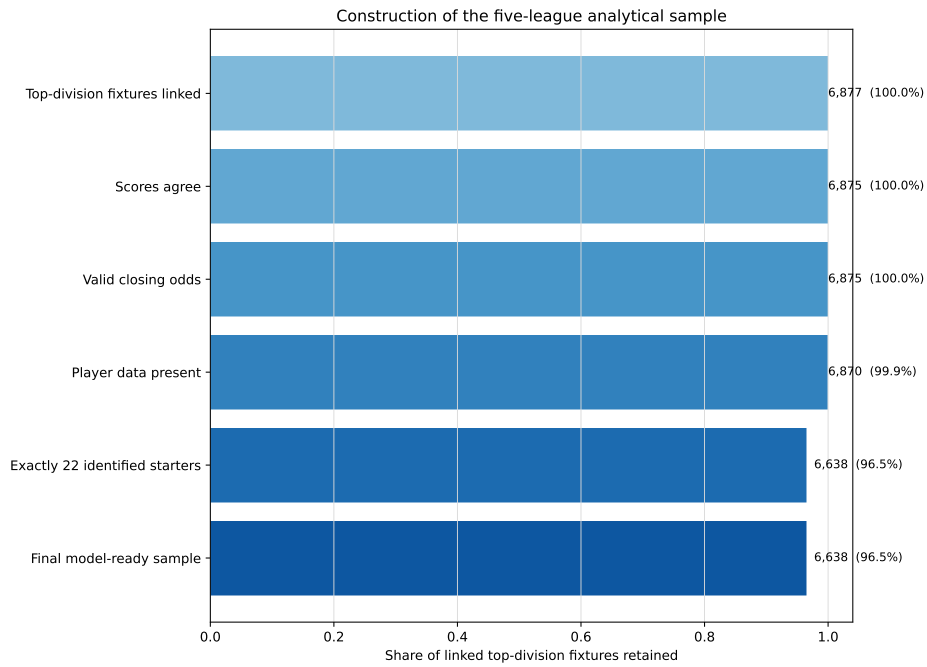
A match was eligible when it was a completed top-division fixture; appeared in both match sources; had identical final scores across them; contained valid closing odds for all three outcomes; contained player data; had eleven starters per side; had complete starter identities and minutes; and contained valid player-to-team assignments. These checks were applied match by match under a single rule set. Each excluded fixture retained a primary reason and its failed checks for audit.

Construction proceeded separately by league. After validating matches, odds, players, and line-ups, pre-match histories were created in date order with the current match excluded. League-prefixed identifiers prevented collisions when the country tables were combined; team histories remained league-specific. The combined data were then split into development and held-out files.

Figure 1 shows how the common rules reduced the linked development sample to the 6,638 matches used for model development and final training. The figure refers to the first five seasons because 2025/26 was kept outside every development decision.

3.5 Data Coverage

The resulting sample contained 7,801 eligible matches: 6,638 from 2020/21–2024/25 and 1,163 from 2025/26. Complete-case selection ensured that all models predicted the same fixtures but reduced representativeness, particularly in Portugal in 2023/24. League counts and the coverage audit are reported in Appendix A.



Retention by league

League	Linked	Final	Retained
Belgium	1,503	1,494	99.4%
Netherlands	1,462	1,459	99.8%
Portugal	1,199	1,000	83.4%
Scotland	1,140	1,121	98.3%
Turkey	1,573	1,564	99.4%

Figure 1: Construction of the five-league development sample. The same eligibility rules were subsequently applied to the held-out 2025/26 season.

4 Methodology

4.1 Prediction Target

The forecasting target was the result of each match after normal time, expressed as one of three mutually exclusive outcomes: home win (H), draw (D), or away win (A). Every method produced a probability for each outcome, and the three probabilities were required to be positive and sum to one. The analysis was therefore concerned with the quality of the complete probability forecast rather than only whether the most likely outcome was correctly identified.

All predictors were defined from information available before the relevant match. Closing odds and announced starting line-ups represented the final pre-match information state. Match results were used only as targets or to update histories for later fixtures. Matches sharing the same kick-off time were processed as a group: their pre-match features were recorded before any result from that group was added to a player or team history.

4.2 Chronological Feature Construction

Player appearances were ordered by kick-off time within each league. For every starter, recent performance was calculated after shifting the appearance history by one match, so the current match could not contribute to its own predictors. The main rolling window contained the player's previous five appearances, with shorter histories retained for new or infrequently observed players. A player's record from another country was not carried into the new league. This rule avoided treating the same recorded performance as directly comparable across competitions with different playing environments and data coverage.

Shots, key passes, and defensive actions—defined as tackles plus interceptions—were converted to rates per 90 minutes before being summarized. For a given starting eleven, the previous-five-appearance rate of each starter was summed to obtain a line-up total. Separate home and away values were first constructed, after which the signed home-minus-away difference was used by the primary logistic models. Thus, for feature z, the match-level value was

$$z_m = z_{m,\text{home}} - z_{m,\text{away}}.$$

The feature-building system also generated alternative windows, exponentially weighted summaries, opponent-adjusted measures, positional summaries, line-up continuity measures, and indicators for limited player history. These variables formed the candidate pool during development; they were not all passed to the final models.

Team-strength variables were updated in the same chronological manner and separately within each league. Each team began with an Elo rating of 1,500, following the standard rating system of Elo (1978). A 60-point home adjustment and a K factor of 20 were used, with the update scaled by the logarithm of the goal margin. Attacking and defensive goal rates began at 1.35 goals and were updated after each match using an exponential weight of 0.20 on the latest observation. A pre-match expected-goals-strength value then combined the team's attacking rate with its opponent's defensive rate; the home value also received a fixed multiplier of 1.10. This is a team-history control, not a player-performance measure.

The primary player-enhanced specification ultimately retained the four home-minus-away variables shown in Table 1. Three describe the recent recorded output of the announced starters, while the fourth controls for recent team scoring strength.

Table 1: Predictors added to the closing-market inputs in the primary player-enhanced model.

Predictor	Construction
Key passes	Sum of each starter's key passes per 90 minutes over his previous five appearances
Shots	Sum of each starter's shots per 90 minutes over his previous five appearances
Defensive actions	Sum of each starter's tackles plus interceptions per 90 minutes over his previous five appearances
Expected-goals strength	Chronological team attacking rate combined with the opponent's defensive rate

Figure 2 summarizes the information flow. The two primary models received the same closing probabilities and were separated only by the selected non-market inputs, while every history was calculated before the match being predicted.

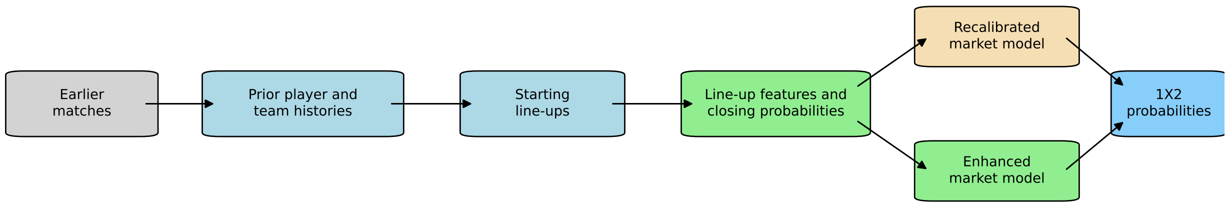


Figure 2: Chronological construction of inputs and the paired primary models.

4.3 Market Probabilities and Models

The inverse of each closing decimal odd was first calculated. The bookmaker margin was removed by dividing each inverse odd by the sum of the three inverse odds. If o_{mk} denotes the closing odd for outcome k in match m, the raw market probability was

$$p_{mk}^{\text{market}} = \frac{1/o_{mk}}{(1/o_{mH}) + (1/o_{mD}) + (1/o_{mA})}.$$

These normalized probabilities formed both a direct benchmark and the inputs to the learned market models. To avoid redundant information, the learned models used two log probability ratios, $\log(p_H/p_D)$ and $\log(p_A/p_D)$.

The primary comparison used two multinomial logistic models with the same estimation procedure. The recalibrated-market model received the two market log ratios and league indicators. It could learn systematic differences between the quoted probabilities and observed outcome frequencies, including league-specific scoring and draw environments. The player-enhanced model received those same inputs plus the four variables in Table 1. Comparing the two therefore tested whether the selected non-market information improved the same market recalibration model. Continuous inputs were standardized using the training sample. Coefficients were estimated with scikit-learn's multinomial logistic regression (Pedregosa et al., 2011), using L_2 regularization ($C = 1$), the limited-memory BFGS solver, and a maximum of 2,000 iterations.

Supporting models broadened the comparison without replacing this primary test. They included historical H/D/A frequencies; a logistic model using non-market features only; and two deterministic LightGBM classifiers (Ke et al., 2017), one based on home-minus-away variables and one retaining separate home and away values. The LightGBM models used 100 trees, a learning rate of 0.03, maximum depth three, seven leaves, a minimum of 30 observations per leaf, and an L_2 penalty of 1.0. Their random seed was fixed at 42.

The score-based benchmark followed Dixon and Coles (1997). Separate attacking and defensive parameters were estimated for each team, together with home advantage and a correction for low-scoring outcomes. Older observations received exponentially declining weight with a half-life of 365 days. Score probabilities from zero to ten goals for each side were calculated and then combined into H/D/A probabilities. A player-enhanced version added a regularized linear shift based on selected home-minus-away player variables to the two teams' expected goal rates. Dixon–Coles models were always estimated separately by league because attack and defence parameters have meaning only within the competition in which the teams played.

Figure 3 summarizes the role of each comparator. The primary test is the paired contrast between the recalibrated market and the same model with the four selected non-market predictors; the remaining models show whether any apparent gain comes from player information, recalibration, or model flexibility.

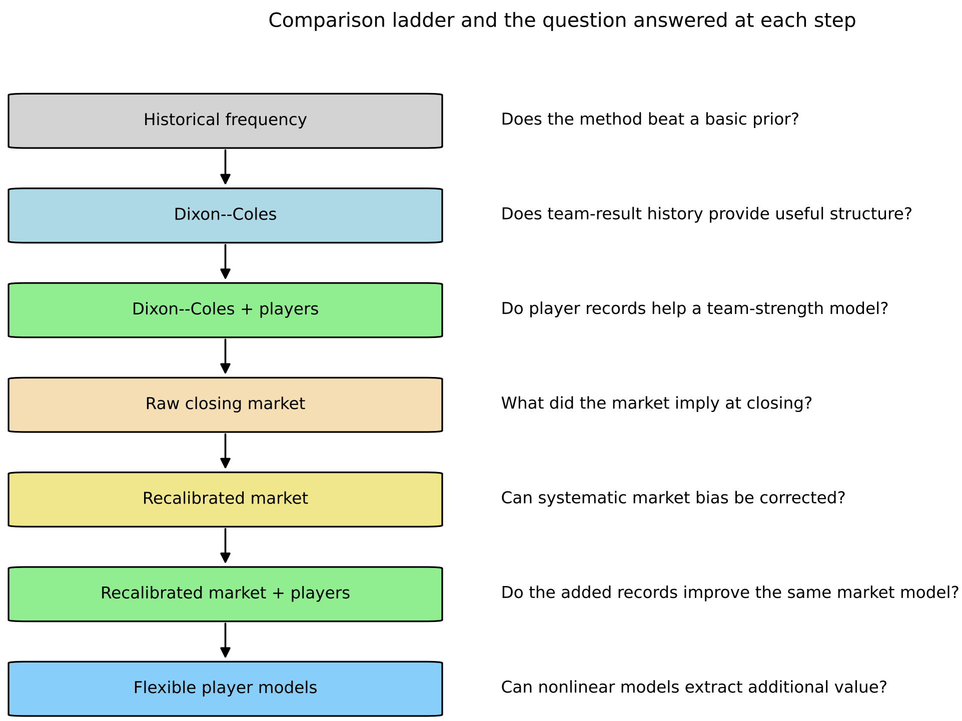


Figure 3: Comparison ladder and the question answered by each model.

4.4 Pooled and League-Specific Training

For models that could share information across countries, the primary training scope pooled all five leagues. League indicators allowed their baseline outcome patterns to differ, while the remaining coefficients were shared. Because the number of eligible fixtures differed by competition, matches were weighted so that every league contributed the same total training weight. If N is the number of training matches, $L = 5$ is the number of leagues, and n_l is the number from league l, each match in that league received weight $N/(Ln_l)$. A larger league could therefore not dominate estimation merely by providing more fixtures.

The same model families, input definitions, and fixed settings were also fitted separately in Belgium, the Netherlands, Portugal, Scotland, and Turkey. These league-specific estimates served as a secondary analysis of whether sharing information was beneficial and whether performance differed across countries. They did not affect the specification selected for the primary pooled test.

4.5 Development Procedure

All modelling decisions were made using the first five seasons. Three expanding-window folds reproduced the information that would have been available at the time of prediction, as shown in Table 2. No later development season was used to fit a model predicting an earlier one.

Table 2: Chronological development folds.

Prediction season	Training seasons
2022/23	2020/21–2021/22
2023/24	2020/21–2022/23
2024/25	2020/21–2023/24

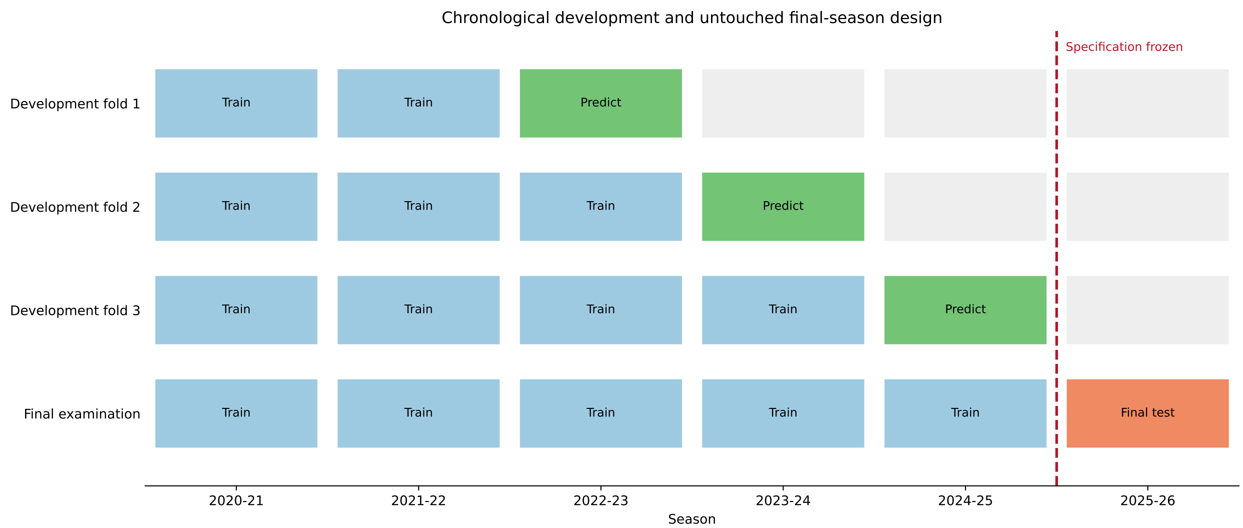


Figure 4: Expanding-window development and the untouched final-season evaluation. The vertical boundary marks the point at which the specification was fixed.

Feature selection was conducted separately for each model family. Candidate variables were first organized into interpretable groups, and all valid group combinations were evaluated through the same three folds. Mean equal-league log loss was the selection criterion. When several combinations were within 0.005 log-loss units of the best result, the smaller set was preferred. The two strongest starting sets were then refined through at most six accepted-search steps. At each step, every permissible one-feature addition and removal was evaluated, and the best move was accepted only if it reduced mean log loss by more than 0.0005. Major feature-block removals, a within-league-and-season player shuffle, and a deliberate future-information test were subsequently used to check that apparent gains were not dependent on one broad input class, arbitrary player assignment, or leakage. The final block-removal check led to removal of the five-appearance mean player rating from the primary market-plus-player model.

The selected feature lists were saved and loaded by both pooled and league-specific evaluators rather than being reselected during later runs. Model settings, feature counts, and a checksum of the saved feature file were also recorded. Tests rejected missing, duplicated, or altered feature entries. Post-hoc probability calibration was examined on development predictions but was not retained because it did not provide a consistent improvement. The final procedure therefore applied no temperature scaling or other adjustment after model fitting.

Figure 5 summarizes the group searches. Each point is an evaluated combination and the line traces the best result at each complexity; vertical values are relative to the best candidate within that model family.

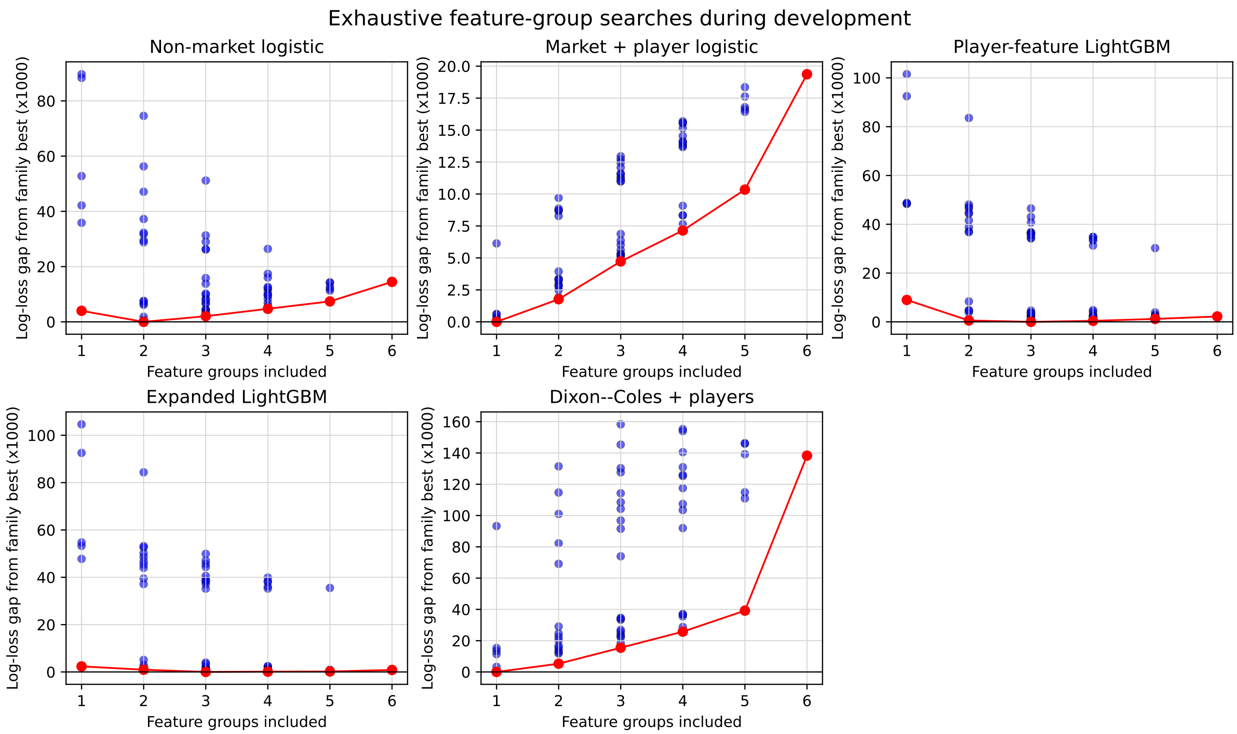


Figure 5: Exhaustive feature-group searches across the five model families. Vertical values are the log-loss gap from the best candidate within the same family, multiplied by 1,000.

4.6 Final Evaluation

After development was complete, the feature lists, model settings, training scopes, comparison rules, scoring measures, league weighting, and uncertainty procedure were fixed in a separate final-evaluation configuration. Each final model was trained once on all eligible matches from 2020/21–2024/25 and then used to predict the previously held-out 2025/26 season. The primary pooled model continued to use equal-league training weights, whereas Dixon–Coles remained league-specific.

Model coefficients were not updated in response to 2025/26 outcomes. However, the chronological player and team histories for a later 2025/26 fixture could naturally include information from earlier fixtures in that season, matching a real forecasting process. Automated checks prevented overlap between the training and final seasons, verified the saved feature checksum, required every model to cover the same eligible matches, and refused to overwrite an existing final-output directory. The final season was not used to revise features, settings, or calibration after its results were observed.

5 Evaluation

5.1 Scoring Measures

Log loss was the primary measure because it evaluates the probability assigned to the observed outcome and strongly penalizes confident errors:

$$\text{LogLoss} = -\frac{1}{N}\sum_{i=1}^{N}\log(p_{i,y_i}),$$

where p_{i,y_i} is the probability assigned to match i's observed outcome. Multiclass Brier score (Brier, 1950) and ranked probability score (RPS) (Epstein, 1969; Murphy, 1971) were secondary measures. RPS compares cumulative probabilities over the ordered H/D/A categories, so probability mass farther from the observed category receives a larger penalty. All three measures are strictly proper scoring rules (Gneiting and Raftery, 2007). Lower values were better; accuracy was descriptive only because it discards probability magnitude.

5.2 League Weighting

The primary summary averaged the five league-level scores, giving each country one fifth of the result regardless of fixture count. Match-weighted and country-specific results were secondary. This evaluation rule mirrored pooled training, where each league also received equal total weight.

5.3 Uncertainty Intervals

Uncertainty was estimated with 10,000 paired bootstrap repetitions, following standard nonparametric bootstrap practice (Efron, 1979; Davison and Hinkley, 1997). Calendar weeks were resampled within league and season, retaining all matches from a selected week and applying the same resample to both models. This preserved some shared temporal conditions and prevented fixture composition from driving model differences. Improvement was defined as

$$\text{improvement} = \text{baseline score} - \text{enhanced-model score}.$$

A positive value favoured the enhanced model. The 2.5th and 97.5th percentiles formed the 95% interval. These intervals quantify variation across the observed fixtures, not uncertainty over alternative data sources or modelling choices.

5.4 Robustness Checks

Development checks removed attacking, defensive, and team-strength feature blocks; jointly shuffled the three player variables within league-season; and introduced an intentionally invalid full-season hindsight variable to verify that leakage would be detected. Temperature scaling was tested only on earlier out-of-sample development predictions and was rejected because it did not improve both primary models consistently. The final claim therefore required a favourable held-out equal-league log-loss difference supported by secondary scores, country results, and the paired intervals.

6 Results

6.1 Development Results

The three chronological development folds produced 3,895 out-of-sample predictions across the five leagues. The enhanced model recorded lower equal-league log loss than the recalibrated market in 2022/23 and 2023/24, but higher log loss in 2024/25. Across all three folds, its log loss was 0.9239, compared with 0.9248 for the recalibrated market and 0.9267 for the raw closing probabilities. The overall development advantage over recalibration was therefore 0.0009 log-loss units, or approximately 0.10%.

Figure 6 makes the change in direction across seasons explicit. The enhanced model improved on recalibration in the first two folds but not in 2024/25, which is why the small pooled development advantage was not treated as settled evidence.

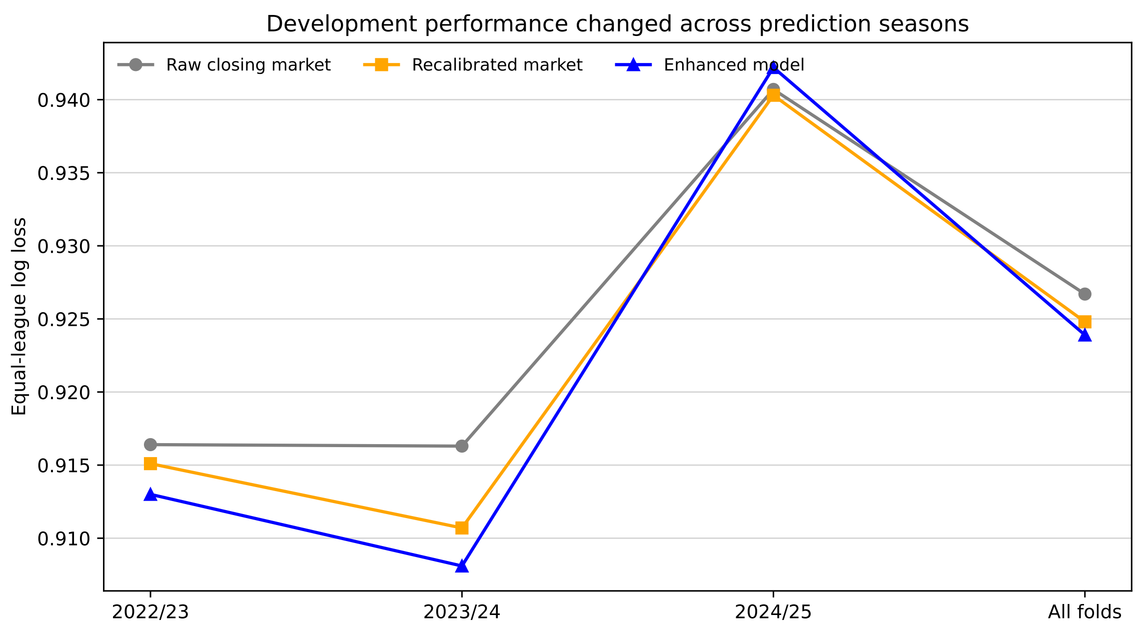


Figure 6: Equal-league log loss across the three chronological prediction seasons and over all development predictions.

Pooling was beneficial during development for the enhanced specification: equal-league log loss was 0.9239 for the shared model and 0.9322 for separate league models. The pooled specification remained the predeclared primary model.

The Dixon–Coles comparisons provided little development evidence of a player effect. The player-enhanced version achieved equal-league log loss of 0.9619, only 0.0002 below the 0.9621 of plain Dixon–Coles. Both were materially worse than the raw and recalibrated closing-market benchmarks. These comparisons were retained as supporting evidence rather than used to select the primary model.

6.2 Final Results

The fixed models were next trained on all 6,638 eligible matches from 2020/21–2024/25 and evaluated on the 1,163 previously held-out matches from 2025/26. Table 3 presents the primary pooled comparison. The enhanced model recorded an equal-league log loss of 0.9580, compared with 0.9567 for the recalibrated market. Because improvement was defined as the baseline score minus the enhanced-model score, the difference was −0.00136, equivalent to a 0.143% deterioration. Brier score and RPS gave the same direction, with relative deteriorations of 0.210% and 0.254%, respectively.

Table 3: Primary pooled comparison on the held-out 2025/26 season. Relative change is positive when the enhanced model is better.

Measure	Recalibrated market	Enhanced model	Relative change
Log loss	0.9567	0.9580	−0.143%
Brier score	0.5697	0.5709	−0.210%
RPS	0.1884	0.1889	−0.254%

The paired match-week interval for the relative log-loss improvement extended from −0.511% to 0.234%. The interval included zero, and 22.9% of the 10,000 resamples favoured the enhanced model. The Brier-score interval ranged from −0.633% to 0.215%, while the RPS interval ranged from −0.828% to 0.327%; both also included zero. The match-weighted estimates were similar. Under that aggregation, the enhanced model deteriorated by 0.154% in log loss, 0.233% in Brier score, and 0.277% in RPS, and none of the three intervals excluded zero.

Figure 7 shows the point estimates and paired match-week intervals together. All six estimates were negative and every interval crossed zero.

Table 4 places the primary result within the wider model suite. Recalibration improved on the raw closing probabilities in both pooled and league-specific training. The pooled non-market logistic and LightGBM models did not match the market-based models. Plain Dixon–Coles remained behind the market, and adding its selected player features increased rather than reduced held-out log loss. Historical outcome frequencies produced the highest log loss of the reported comparators.

The league-specific enhanced market model had the lowest descriptive log loss in the full table, 0.9542, compared with 0.9561 for league-specific recalibration. Its Brier score was also lower, at 0.5681 against 0.5688, whereas its RPS was fractionally higher, at 0.1880 against 0.1880 before rounding to four decimal places. This secondary result arose from five separately estimated models and did not replace the predeclared pooled comparison.

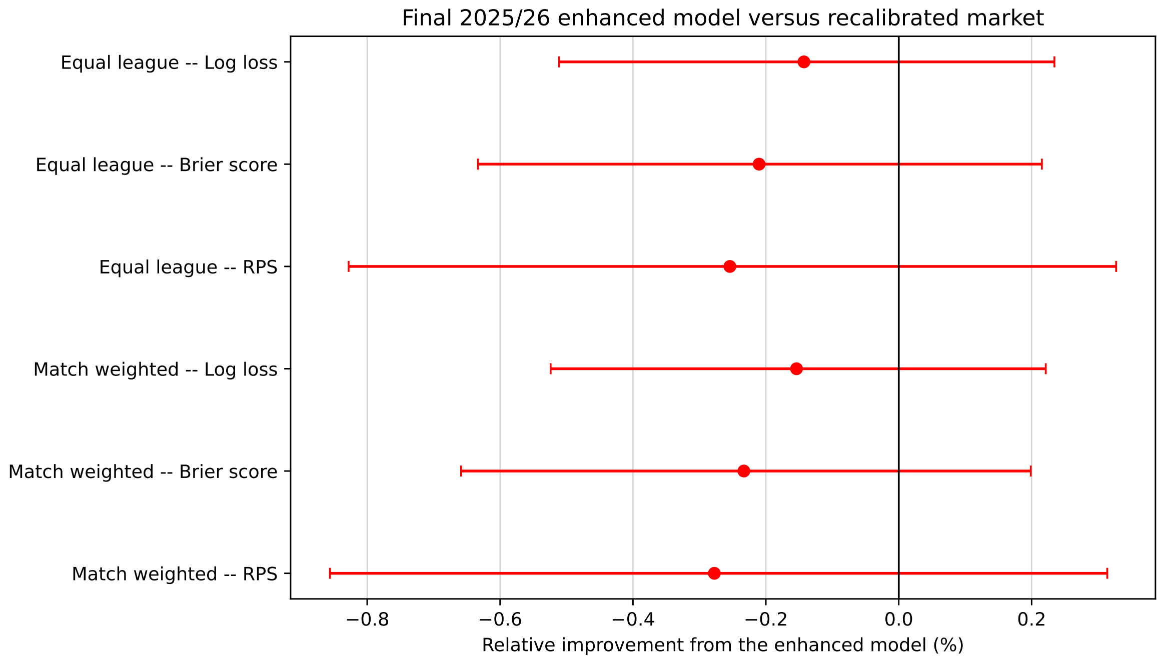


Figure 7: Relative improvement of the pooled enhanced model over the pooled recalibrated market in 2025/26. Positive values favour the enhanced model; bars are paired 95% match-week bootstrap intervals.

6.3 Country-Level Results

The pooled primary comparison varied in direction across countries (Table 5). The enhanced model reduced log loss in the Netherlands and Portugal but increased it in Belgium, Scotland, and Turkey. The largest point improvement occurred in Portugal, at 0.450%, while the largest deterioration occurred in Belgium, at 0.629%.

Every country-level interval included zero. The proportion of resamples favouring the enhanced model ranged from 5.5% in Belgium to 84.5% in Portugal, with values of 60.8% in the Netherlands, 20.8% in Scotland, and 31.1% in Turkey. The country results therefore showed mixed directions without an interval that clearly separated the two models.

6.4 Robustness Results

All robustness decisions were made before the holdout was opened. Removing mean player rating improved the five-feature precursor from 0.9241 to 0.9239, yielding the final four-feature model. Other block removals changed loss by less than 0.0006, while shuffling players worsened it from 0.9241 to 0.9250. The invalid hindsight feature produced the expected artificial gain (0.8928), confirming the importance of chronological construction. Temperature scaling worsened both primary models and was not retained. Full development diagnostics appear in Appendix D.

Table 4: Equal-league log loss of the principal comparators in 2025/26.

Training scope	Model	Log loss
League-specific	Enhanced market model	0.9542
League-specific	Recalibrated market	0.9561
Pooled	Recalibrated market	0.9567
Pooled	Enhanced market model	0.9580
Pooled/league-specific	Raw closing market	0.9603
Pooled	Player-feature LightGBM	0.9720
Pooled	Non-market logistic model	0.9758
League-specific	Dixon–Coles	0.9810
League-specific	Dixon–Coles with players	0.9820
Pooled/league-specific	Historical frequency	1.0790

Table 5: Pooled primary log-loss comparison by league in 2025/26. Positive relative change favours the enhanced model.

League	Market	Enhanced	Relative change	95% interval
Belgium	0.9930	0.9992	−0.629%	−1.386%, 0.137%
Netherlands	0.9843	0.9833	0.109%	−0.684%, 0.845%
Portugal	0.9074	0.9033	0.450%	−0.406%, 1.314%
Scotland	0.9596	0.9633	−0.383%	−1.310%, 0.528%
Turkey	0.9391	0.9412	−0.220%	−1.059%, 0.671%

7 Discussion

7.1 Interpretation and Practical Implications

The study found no reliable evidence that the tested recent player-performance variables improved pooled match forecasts beyond recalibrated closing probabilities. A small development advantage reversed in the untouched 2025/26 season: all three proper scores favoured the market-only model slightly, their paired intervals crossed zero, and country estimates had mixed signs. This is evidence against a dependable gain from the tested representation, not against player information in general.

One likely explanation is that closing odds already reflected much of the information represented by the added variables. A variable can describe genuine team differences yet add nothing conditional on prices that already incorporate line-ups, injuries, transfers, recent form, and other market information. The features were also noisy: five-appearance event rates mix ability with opponent, role, tactics, score state, and playing time, while line-up totals discard partnerships and positional interactions.

The difference between development and final performance is itself informative. It was small, present in only two of three folds, and selected after comparing many candidates. Its reversal illustrates why chronological development alone does not replace an untouched test. It also shows why the benchmark must match the claim: recalibration improved raw prices, but the added predictors did not improve the same recalibrated model.

For applied forecasting, the added features also carry a practical cost. They require collection and reconciliation of line-ups, player identities, minutes, positions, and event statistics, followed by chronological updating for every player. In the present setting, this extra data and modelling burden was not matched by a measurable held-out improvement. There is therefore no empirical reason to prefer the enhanced pooled model when reliable closing probabilities are available.

Greater flexibility did not solve the problem. The LightGBM specifications performed worse than the primary logistic models, and adding player features to Dixon–Coles also failed on the holdout. In this application, additional complexity did not compensate for weak incremental signal.

The negative result remains useful. It rules out a plausible but costly feature design under a demanding evaluation and prevents a small development gain from being presented as a general improvement. It also provides a fixed standard for testing richer player representations against the same market-only baseline.

7.2 Heterogeneity and Relation to Previous Research

The country-level point estimates were heterogeneous. Portugal and the Netherlands favoured the enhanced pooled model, whereas Belgium, Scotland, and Turkey favoured recalibration alone, but every country interval crossed zero. The slight advantage of separately trained enhanced models is likewise hypothesis-generating: it was secondary, used five more flexible fits, and was not consistent across scores. Additional seasons under an unchanged specification are needed before attributing effects to particular leagues.

The strong closing benchmark agrees with earlier evidence that bookmaker odds are difficult to outperform (Forrest et al., 2005; Štrumbelj and Robnik-Šikonja, 2010). The result also reinforces the importance of the raw-versus-recalibrated distinction: odds conversion and market efficiency can affect assessment (Štrumbelj, 2014; Angelini and De Angelis, 2019), so comparison with raw probabilities alone would confound player information with calibration.

The findings do not contradict studies showing that player ratings improve team-based models (Arntzen and Hvattum, 2021). Those studies ask whether players help team histories; this study asks whether recent observed output adds value after conditioning on an information-aggregating market. Its weak contribution is closer to Gelade and Hvattum (2020), while the latent ability model of Holmes and McHale (2024) addresses a different, potentially more stable player representation.

8 Limitations

First, the tested representation was narrow. Five-appearance key-pass, shot, and defensive-action rates mix ability with opponent, role, tactics, score state, and playing time; line-up totals capture partnerships and positional balance only indirectly. Histories were also restricted to the current league, discarding potentially useful information after transfers.

Second, the primary enhanced model added three player variables and one team-strength control. The holdout therefore identifies the contribution of the four-feature bundle, not a player-only model. Development ablations help with interpretation but cannot replace a predeclared player-only holdout comparison.

Third, generalization is limited by one final season, five competitions, and complete-case selection. Country samples of 184–267 matches produced wide intervals; Greece was excluded for line-up coverage, and fixtures with missing odds or incomplete player records may differ systematically from those retained.

Finally, closing odds may already incorporate player information that the study cannot identify separately. The model suite also excludes dynamic player ability, tactical and interaction models, and explicit absence models. The results concern probability forecasts, not betting returns: no available-price, limit, execution, or staking process was evaluated. These boundaries restrict the claim to the tested summaries and design rather than player information in general.

9 Conclusion

This study tested whether recent performances of announced starters improved three-way forecasts beyond recalibrated closing odds. All decisions were made on 6,638 matches from 2020/21–2024/25 before evaluation on 1,163 held-out 2025/26 matches across five leagues.

The development advantage did not generalize. Equal-league log loss was 0.9580 for the enhanced model and 0.9567 for recalibration alone, a 0.143% deterioration; Brier score and RPS agreed, and every country interval crossed zero. The tested combination of recent starter summaries and team-strength context therefore showed no reliable incremental value.

This does not imply that all player information is uninformative. It shows that added data and complexity require comparison with an equally trained market-only baseline on unseen data. Richer player representations remain plausible, but they should retain chronological construction, identical-match comparisons, and an untouched final evaluation.

AI Usage Disclosure

During the preparation of this work, the author used ChatGPT within the Codex environment to assist with selected coding tasks, code documentation, and streamlining the development workflow. Grammarly was subsequently used to review the manuscript for grammar, spelling, and writing style. All outputs generated with the assistance of these tools were critically reviewed, verified, and edited by the author. The author takes full responsibility for the accuracy, integrity, and final content of the manuscript and the accompanying code.

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A Data Quality and Sample Coverage

The common eligibility procedure was applied independently inside each league before the country tables were combined. A match entered the modelling sample only when it was a completed top-division fixture, appeared in both match sources with an agreeing score, had valid closing prices for all three outcomes, contained player records, and supplied exactly eleven identified starters for each team with valid minutes and team assignments. The same requirements were used in development and in the held-out season.

Table 6: Eligible observations used for final training and evaluation.

League	2020/21–2024/25	2025/26	Total
Belgium	1,494	267	1,761
Netherlands	1,459	255	1,714
Portugal	1,000	211	1,211
Scotland	1,121	184	1,305
Turkey	1,564	246	1,810
Total	6,638	1,163	7,801

Greece was not included because its starting-line-up coverage in the first two seasons failed the declared coverage requirement. The exclusion was made before the five-league model comparisons and was not determined by forecasting performance. Portugal's reduced 2023/24 sample was retained under the common rules rather than replaced by a country-specific exception.

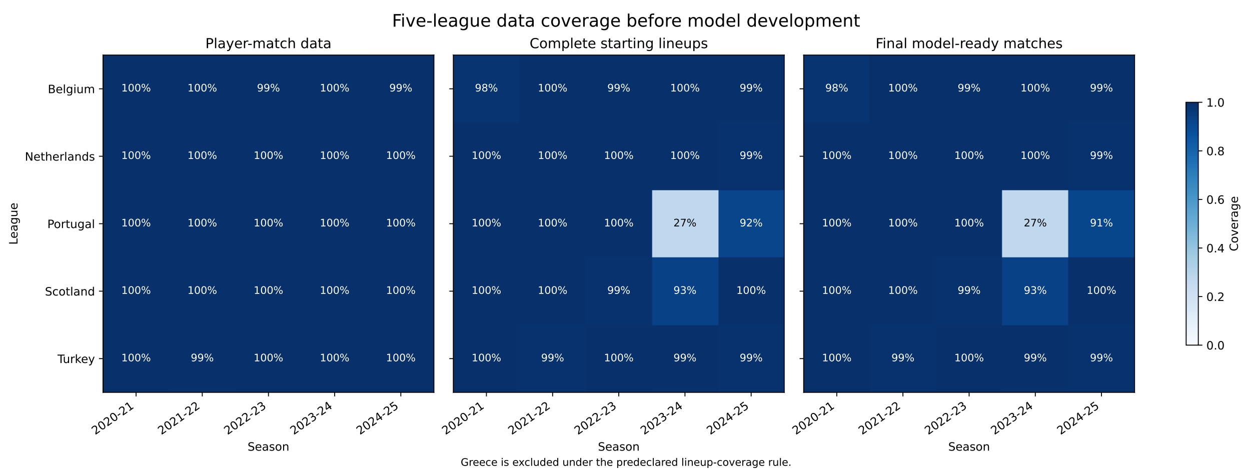

Figure 8: Development-period data coverage by league and season. Percentages are relative to linked top-division fixtures in each league-season.

B Full Model Specifications

Table 7: Information and estimation scope of the evaluated models.

Model	Training scope	Information used
Historical frequency	Shared and separate	Earlier league-specific H/D/A frequencies
Raw closing market	None	Margin-removed closing home, draw, and away prices
Recalibrated market	Shared and separate	Closing-market log ratios and league indicators
Non-market logistic	Shared and separate	Selected chronological player and team variables, without odds
Enhanced market logistic	Shared and separate	Recalibrated-market inputs plus four selected non-market variables
Player-feature LightGBM	Shared and separate	Selected home-minus-away player and team variables
Expanded LightGBM	Shared and separate	Selected home and away variables retained separately
Dixon–Coles	Separate by league	Earlier scores, time-weighted team attack and defence, and home advantage
Dixon–Coles with players	Separate by league	Dixon–Coles structure plus selected announced-line-up variables

Table 8: Fixed estimation and evaluation settings.

Component	Fixed setting
Logistic regression	Standardized inputs; L_2 regularization with $C = 1$; limited-memory BFGS; maximum 2,000 iterations
LightGBM	100 trees; learning rate 0.03; maximum depth 3; 7 leaves; minimum 30 observations per leaf; L_2 penalty 1.0
Shared training	Each league assigned equal total training weight
Dixon–Coles	Separate model per league; older matches exponentially downweighted with a 365-day half-life
Model seed	42
Primary score	Equal-league log loss
Secondary scores	Brier score and ranked probability score
Uncertainty	10,000 paired match-week bootstrap repetitions; seed 42
Post-hoc calibration	None in the final evaluation

C Selected Features

The entries below are the exact names loaded by the final evaluators. They were selected entirely from the chronological development folds. The shared and separate versions of a model used the same feature list so that training scope, rather than input availability, was the relevant difference.

Table 9: Final selected features by model family.

Model family	Selected feature
Non-market logistic	`elo_rating`
	`opponent_elo_rating`
	`elo_difference`
	`defence_goal_rate_ewm`
	`expected_goals_strength`
	`key_passes_per90_sum_5`
	`adjusted_defensive_actions_lineup_mean_5`
	`adjusted_key_passes_lineup_std_across_starters_5`
Enhanced market logistic	`key_passes_per90_sum_5`
	`shots_per90_sum_5`
	`defensive_actions_per90_sum_5`
	`expected_goals_strength`
Player-feature LightGBM	`key_passes_per90_sum_5`
	`shots_per90_sum_5`
	`defensive_actions_per90_sum_5`
	`recent_minutes_sum_5`
	`rating_mean_5`
	`starters_without_history`
	`starters_without_full_window`
	`elo_rating`
	`opponent_elo_rating`
	`elo_difference`
	`elo_expected_result`
	`prior_team_matches`
	`attack_goal_rate_ewm`
	`defence_goal_rate_ewm`
	`expected_goals_strength`
	`key_passes_per90_lineup_mean_ewm`
	`adjusted_defensive_actions_lineup_mean_1`
	`rating_lineup_mean_3`
	`adjusted_rating_lineup_std_across_starters_5`
Expanded LightGBM	`elo_rating`
	`opponent_elo_rating`
	`elo_difference`
	`elo_expected_result`
	`prior_team_matches`

Model family	Selected feature
	attack_goal_rate_ewm key_passes_per90_sum_5 adjusted_key_passes_lineup_mean_1 gk_saves_per90_5 adjusted_key_passes_lineup_mean_ewm
Dixon–Coles with players	key_passes_per90_sum_5 recent_minutes_sum_5 rating_mean_5 starters_without_history mid_rating_mean_5 fwd_adjusted_shots_5 gk_saves_per90_5

D Additional Development and Diagnostic Results

Figure 9 reports the focused development checks relative to the five-feature precursor to the primary model. Negative values mean that the alternative reduced log loss. Removing the mean player rating was the only tested removal that improved the precursor and therefore produced the four-feature final specification. The remaining feature-block removals, the market-only comparison, and the player shuffle all increased loss, although the changes were small.

Figure 10 compares development-period probability reliability for the raw closing market and the shared and separately trained enhanced models. It is diagnostic rather than the primary evaluation: the final research claim rests on proper scoring rules and the paired comparison against the recalibrated market.

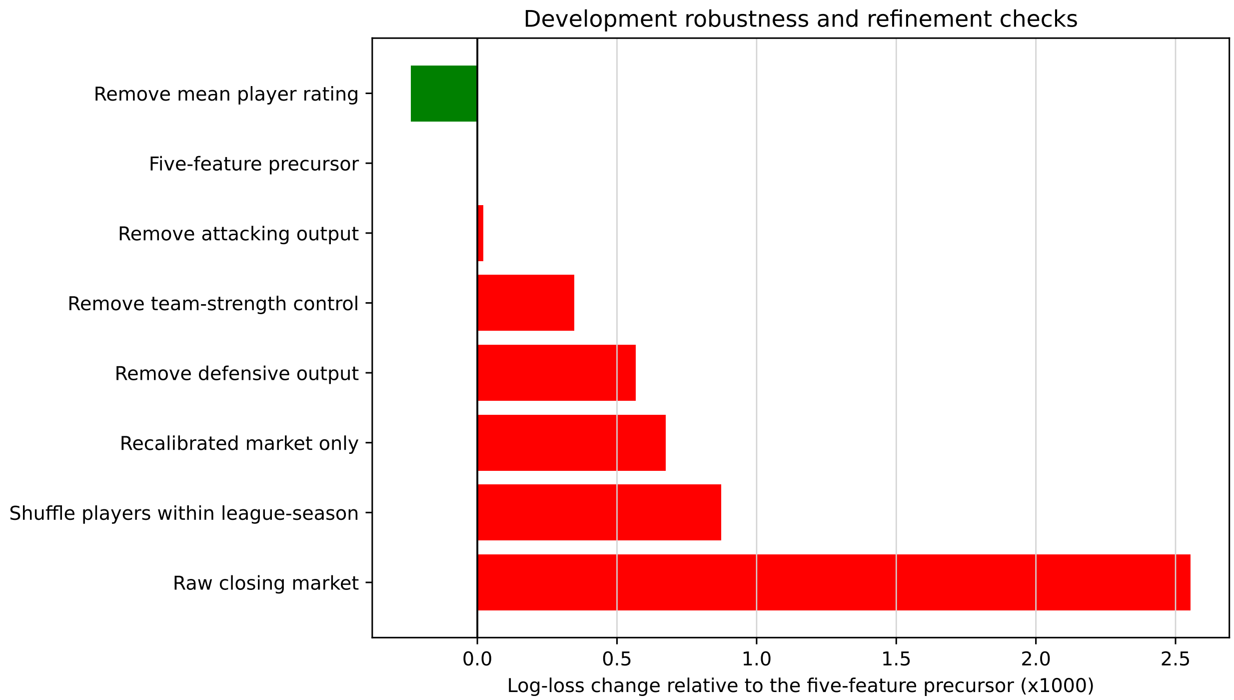

Figure 9: Development robustness checks relative to the five-feature precursor. Values are log-loss changes multiplied by 1,000.

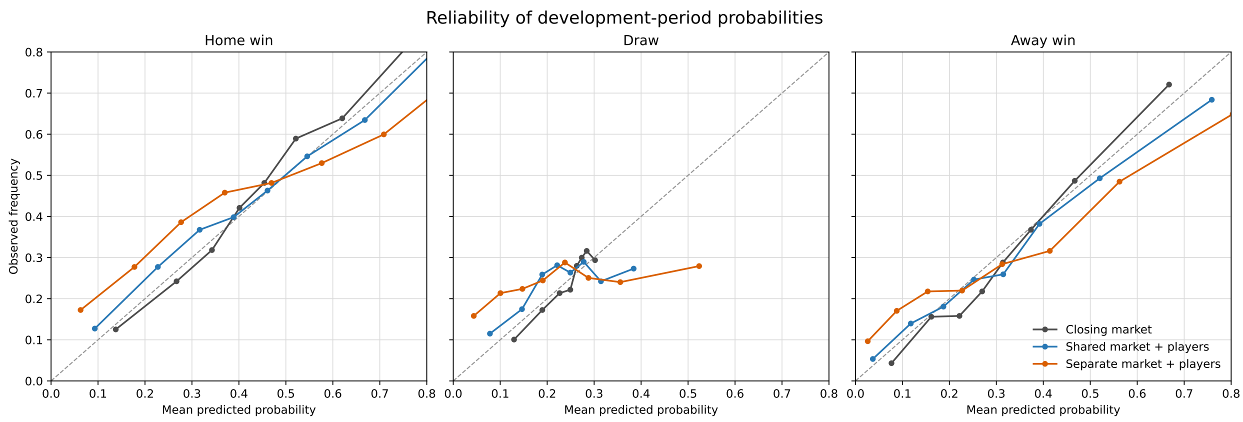

Figure 10: Development-period reliability by outcome. Points compare mean predicted probabilities with observed frequencies in probability bins.